

Stochastic Life Science Simulation Framework

Event-driven modeling of complex biological systems using Gillespie sampling

Python · Computational Biology · Mathematical Modeling
Reusable methodology for stochastic biological processes

● Reproducible simulation design

SIMULATION METHOD

Direct Gillespie

Continuous-time stochastic events

SYSTEM BEHAVIOR

Fixed + dynamic

Static and evolving structures

REPRODUCIBILITY

Seeded ensembles

Repeatable random experiments

COMPUTATION

Parallel runs

Independent simulations across CPUs

SCIENTIFIC MODELING PIPELINE

01

Biological mechanism

Define molecular interactions, physical constraints and experimentally observed behaviors.



02

Model parameters

Convert rates, forces, diffusion and geometry into named configurable inputs.



03

Event system

Represent attachment, detachment, movement, growth and state changes as explicit events.



04

Stochastic engine

Sample next-event time and identity from current state-dependent propensities.



05

Ensemble analysis

Aggregate independent traces into distributions, uncertainty and condition comparisons.

State-dependent rates

Event probabilities change as biological state evolves.

Random waiting time

Continuous-time delay sampled from total propensity.

Weighted event choice

Next mechanism selected according to relative rates.

Iterative state update

Every event becomes input for next simulation step.

Molecular kinetics

Binding, unbinding and reaction-rate processes.

Cellular mechanics

Force-dependent behavior and physical constraints.

Dynamic structures

Growth, shrinkage, transitions and moving boundaries.

Condition comparison

Control versus perturbed biological scenarios.

GILLESPIE SIMULATION CYCLE

Calculate rates

Read current biological state.



Sample time

Draw stochastic waiting interval.



Select event

Use cumulative propensities.



Update state

Apply mechanism and constraints.



METHODOLOGICAL STRENGTHS



Mechanism-first design

Named events make biological assumptions visible and auditable.



Analytical connection

Model equations and stochastic simulation share one parameter source.



Reproducible randomness

Master seeds create unique, repeatable traces across workers.



Parallel ensembles

Independent traces scale across CPU processes without changing event order.



Scientific safeguards

Boundary checks preserve physical and biological constraints.



Transparent assumptions

Unavailable measurements and substitute parameters remain explicitly documented.



Analysis-ready outputs

Traces, categories, means and uncertainty summaries support reporting.